Systematic Market-Neutral Crypto: A Readable Statistical Signal (Sharpe 3.45)

By Talha Çağrı Kotcioğlu — Data Scientist, Crypto Quant · LinkedIn post

I'd like to share my newly deployed model with you.

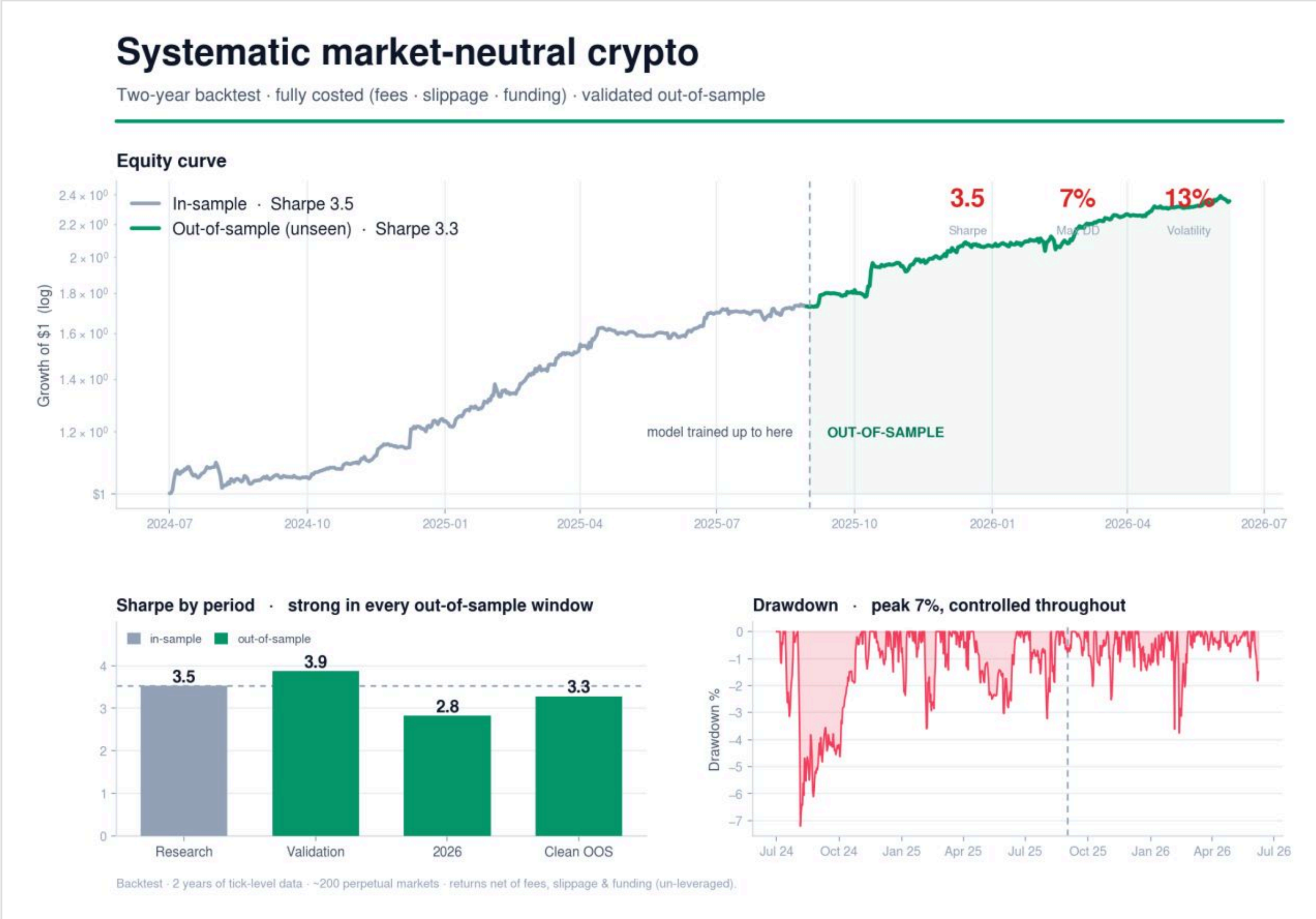
- Sharpe ratio: 3.45
- Sortino ratio: 4.42
- Profit factor: 1.80
- Max drawdown: 7.20%

Market-neutral crypto. 2 years of tick-level data. Every cost baked in.

It's not HFT, resistant to 1 hour execution delay. No ML. No black box. A statistical signal you can read top to bottom. No leverage. Taker fees, conservative slippage and funding, all included. Long and short, so it doesn't need the market to go up. It rode through every major liquidation cascade in the window and still capped out at a 7% drawdown.

My only concern is capacity, so I've deployed it live to measure how it holds up.

#Crypto #QuantFinance #AlgoTrading #QuantitativeTrading



Author's chart: two-year backtest, fully costed (fees + slippage + funding), validated out-of-sample. Equity curve (log): in-sample Sharpe 3.5, out-of-sample Sharpe 3.3. Sharpe by period — Research 3.5, Validation 3.9, 2026 2.8, Clean OOS 3.3. Drawdown: peak ~7%, controlled throughout.

From the comment discussion

Asked how it handles flash crashes and sudden moves, the author explained the core mechanism:

> **Talha Çağrı Kotcioğlu (author):** "Definitely the best advantage of this strategy is pre-sizing and parity choosing. Before crashes, it sizes heavily on shorts. The disadvantage is, when the market is too flat, the strategy is losing 1-2%. So, the crashes/explosions are source of pnl in this strategy."

On capacity and execution:

> **Alex Koval:** "If it truly works up to 1 hour execution, then you could just TWAP over the hour and should have no capacity concerns on main coins."

> **Talha Çağrı Kotcioğlu (author):** "Definitely there is no problem on main coins. Even if I target >$10M daily volume, the portfolio includes some alt coins as well. Those are a little problematic."

> **Talha Çağrı Kotcioğlu (author):** "I deployed it 3 weeks before, everything goes as expected. I am increasing the size step by step to monitor capacity."

Skeptical pushback centred on the short sample:

> **Kyrylo Maltsev (Head of Quant BD, TradeGreat):** "Backtest with period of 2 years is absolutely useless... Please don't jump in this trap with real money."

> **Marvin Appiah-kubi:** "You captured 2 years in sample which isn't even half of a full market cycle and your out of sample is 6 months... Your model is very probable to cave in under the pressure of a completely unforeseen regime."

> **Mikko Ohtamaa:** "If you backtest on a new exchange like Hyperliquid you effectively capture its 2024-2025 bull market and then your model falls apart in 2026."

Source: https://www.linkedin.com/feed/update/urn:li:activity:7475257732896309248/